# Human Computer Interaction

Sec 1



- Human-Computer Interaction (HCI) is the field that studies how people interact with computer systems and how to design those systems to be usable, efficient, and satisfying.
- The goal is not only to make the system work, but to make it work well for humans.

# Why is HCI Important?

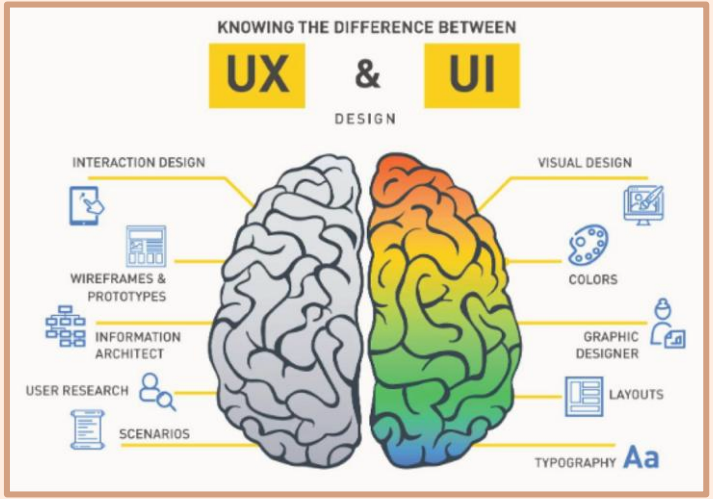
- Many systems fail not because of bad code, but because of poor design.
- Bad user experience leads to user leave.
- It reduces human errors.
- It increases productivity.

A technically perfect system can still fail if users cannot understand how to use it.

## UX (User Experience)

The overall feeling and experience while using the system.

- Is it easy?
- Is it frustrating?
- Can users achieve their goals quickly?



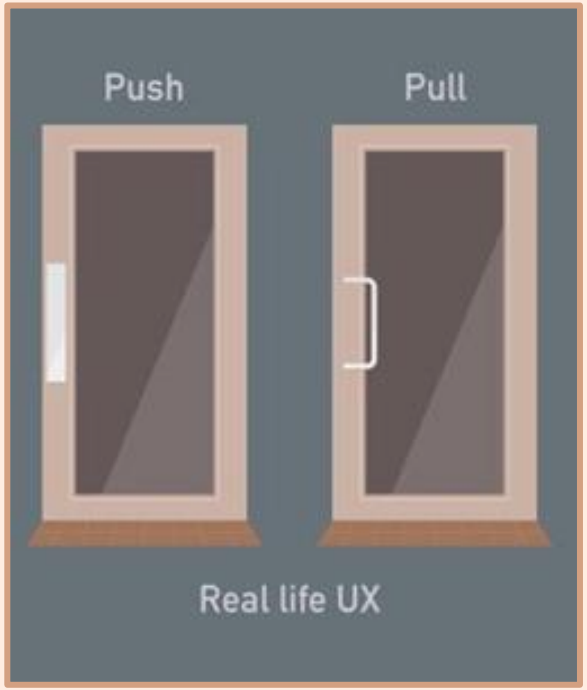

## UI (User Interface)

The visual part of the system.

- Colors
- Buttons
- Typography
- Layout

It is what users see.

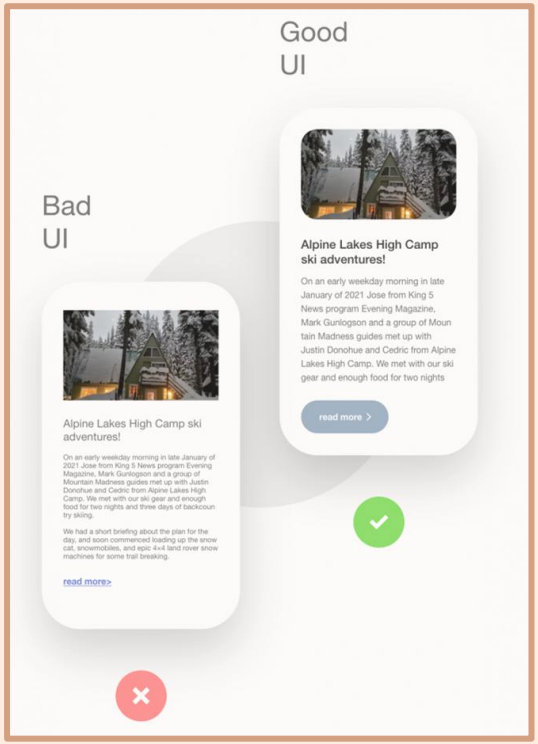
# Real life example

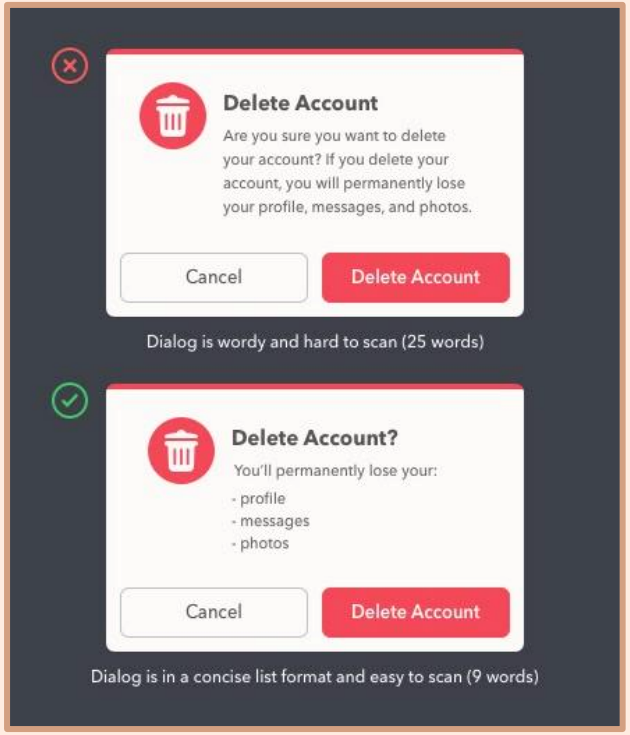




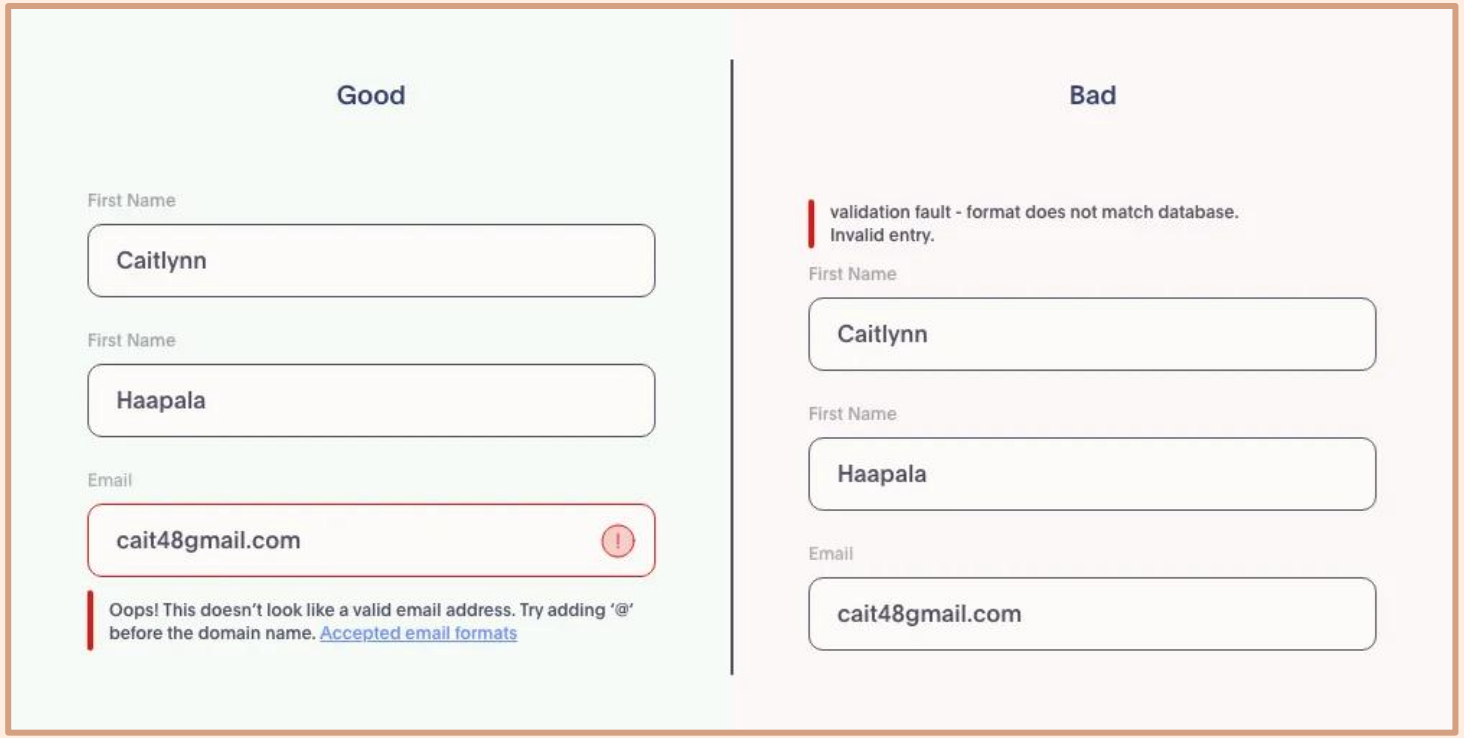



# Examples

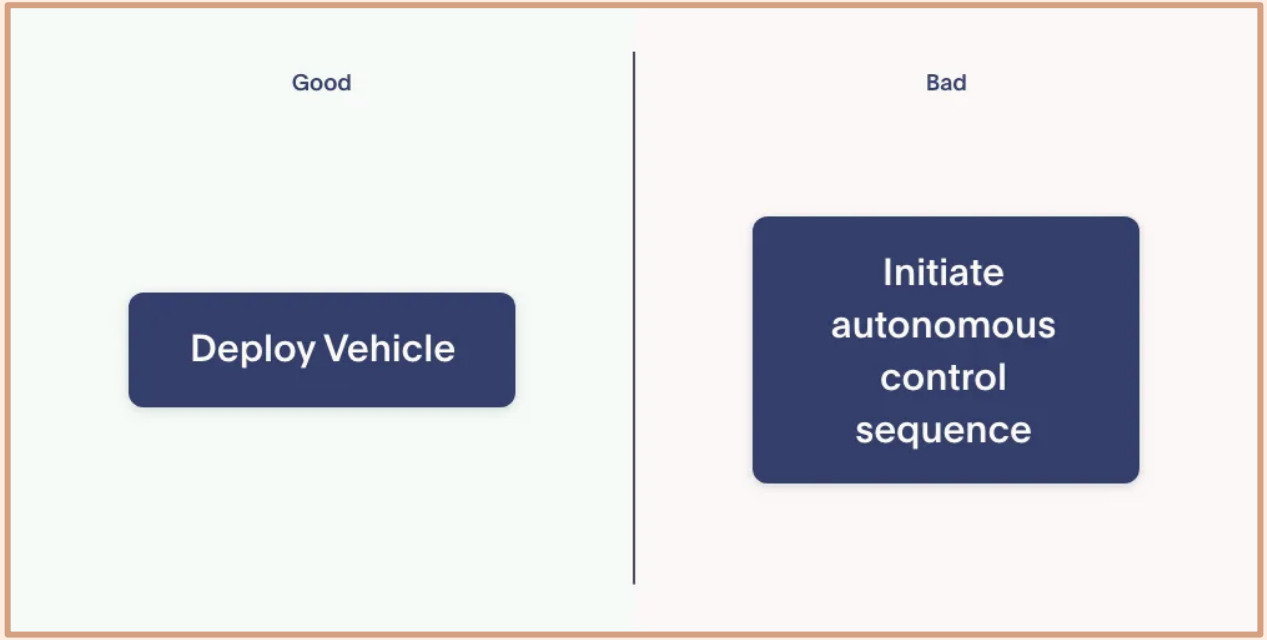

Good
UI
Bad
UI
Alpine Lakes High Camp ski adventures!
On an early weekday morning in late January of 2021 Jose from King 5 News program Evening Magazine, Mark Gunlogson and a group of Mountain Madness guides met up with Justin Donohue and Cedric from Alpine Lakes High Camp. We met with our ski gear and enough food for two nights and three days of backcoun try skiing.
We had a short briefing about the plan for the day, and soon commenced loading up the snow cat, snowmobiles, and epic 4×4 land rover snow machines for some trail breaking.
read more>
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read more >


Delete Account
Are you sure you want to delete your account? If you delete your account, you will permanently lose your profile, messages, and photos.
Cancel
Delete Account
Dialog is wordy and hard to scan (25 words)
Delete Account?
You'll permanently lose your:
- profile
- messages
- photos
Cancel
Delete Account
Dialog is in a concise list format and easy to scan (9 words)

# Examples

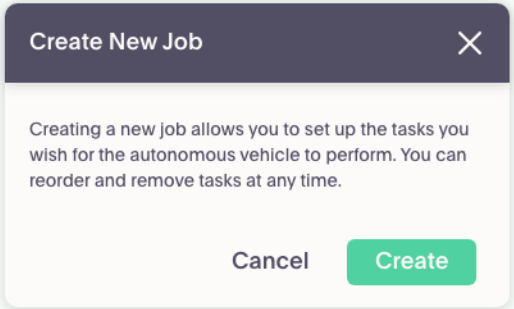

Good
First Name
Caitlynn
First Name
Haapala
Email
cait48gmail.com
Oops! This doesn't look like a valid email address. Try adding '@' before the domain name. Accepted email formats
Bad
validation fault - format does not match database.
Invalid entry.
First Name
Caitlynn
First Name
Haapala
Email
cait48gmail.com

# Examples


Good
Deploy Vehicle
Bad
Initiate autonomous control sequence

# Examples

Good

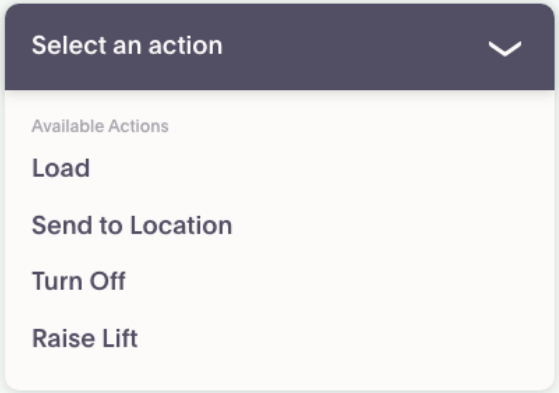



Bad

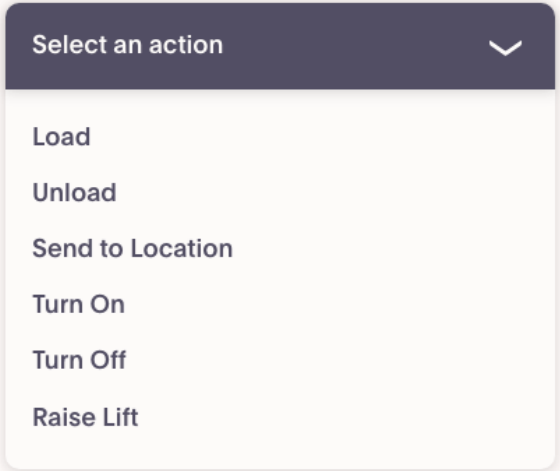
Create New Job: Creating a new job allows you to set up the tasks you wish for the autonomous vehicle to perform. You can reorder and remove tasks at any time. **Cancel / Create**

# Examples

Good




Bad



# Components of an Interactive System

01 Human

02 Computer

03 The Interaction

# 01 Human

We study human capabilities and limitations.

## 🧠 Perception

How humans interpret: Colors, Sounds, Movement

→ Example:

Red = danger or warning

Green = success or confirmation

If colors are misused → confusion occurs.

## 🧠 Memory

- Short-term memory (very limited)
- Long-term memory

→ Short-term memory can hold only about 7±2 items.

→ Therefore:

- Do not overload users with too many options at once.
- Do not require users to remember complex information.

# 01 Human

## 🧠 Attention

Humans cannot focus on many things at the same time.

→ If a page contains:

Too many animations

Many colors

Pop-ups

Ads

It causes cognitive overload.

**Cognitive Load:** is the mental effort required to use the system.

# 02 Computer

- Hardware (keyboard, mouse, touchscreen)
- Software
- System response time

→ Example:
If a user clicks a button and nothing happens,
they may think the system is broken.

→ System responsiveness is part of good interaction.

# 03 The Interaction

Interaction consists of:

- **Input**
  - Typing
  - Touch
  - Voice
  - Mouse clicks
- **Output**
  - Visual display
  - Audio feedback
  - Notifications
- **Feedback (Very Important)**
  - Every action must have a visible reaction.

→ Example:
Clicking “Submit” → show loading indicator.
Successful submission → show confirmation message.

Without feedback, users feel lost.

# Design Principles

## 1. Visibility

Users should clearly see available options.

→ Example: The “Delete” button should not be hidden in an unclear location.

## 2. Consistency

Similar elements should behave the same way across the system.

→ Example: If the “Save” button is blue everywhere, it should not suddenly become green.

## 3. Feedback

The system must respond to user actions.

→Example: Uploading a file → show a progress bar.

# Design Principles

## 4. Affordance

The design should suggest how to use an object.

→ Example: A raised button suggests it can be clicked , Underlined blue text suggests a hyperlink.

**Users should not need instructions to understand basic elements.**

## 5. Simplicity

The best design is often the simplest one.

→ Example: Google's homepage
Very simple, yet highly effective.

## 6. Error Prevention

Good design prevents errors before they happen.

→ Example: If a national ID requires 14 digits,
the system should prevent entering fewer digits.

**Prevention is better than correction.**

## Usability

A measure of how easy and effective the system is to use.

Usability is the degree to which a system can be used effectively, efficiently, and satisfactorily to achieve specific goals.



# The Five Usability Attributes

**1. Learnability**
How easy is it for first-time users to accomplish tasks?

**2. Efficiency**
Once users learn the system, how quickly can they perform tasks?

**3. Memorability**
If users return after a long time, can they use the system without relearning everything?

**4. Errors**
How many errors do users make?
How serious are these errors?
Can they easily recover?

**5. Satisfaction**
Do users enjoy using the system?
Is the experience pleasant?

# THANKS

Eng/ Hajar Nagm